

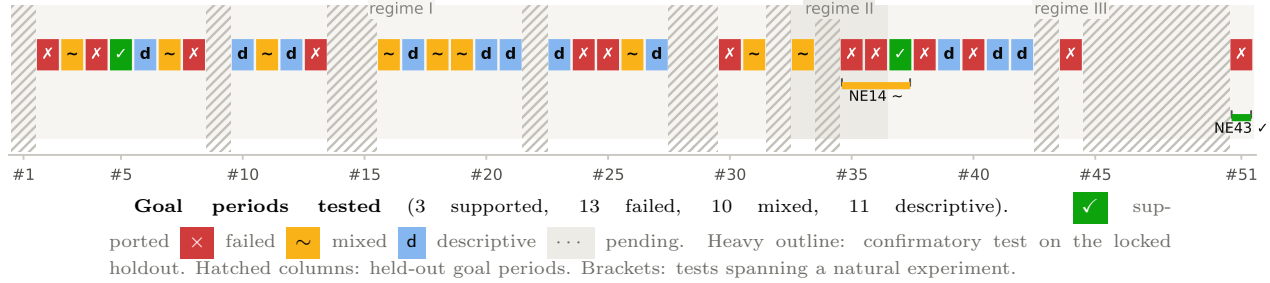
# H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

**Bottom line.** The swarm is clearly subcritical ( $n \approx 0.2\text{--}0.4$ ) and its self-excitation does not depend on the goal's coupling mode; only a small, fast, reply-like cross-triggering survives its null.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



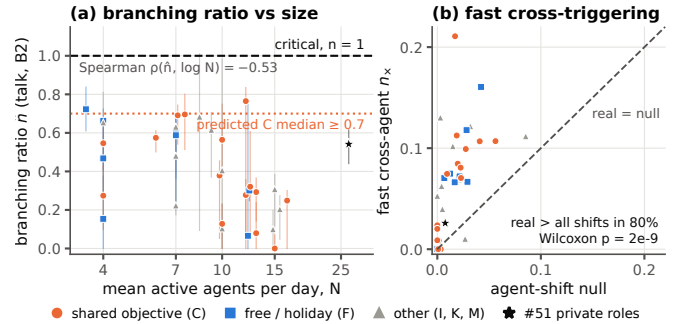
## WHAT HAPPENED

Hawkes fits to agent talk events in 35 non-holdout goal periods (282 days), each fitted separately with a schedule baseline. Predicted: shared-objective weeks run near critical (median  $\hat{n} \geq 0.7$ ) and above free weeks, and #51 drifts toward  $n = 1$  as its roster grows. All failed. Median  $\hat{n}$  is 0.41 (0.36 in round 1b), bracketed at 0.2–0.4 by the baseline choice. Shared-objective weeks gave 0.32 vs 0.53 for free weeks ( $p = 0.75$ );  $\hat{n}$  falls with swarm size ( $\rho = -0.53$ ), and in #51 as its roster grows ( $\rho = -0.64$ ). A constant baseline would have “confirmed” criticality ( $\hat{n} \approx 0.99$ ), and pooled  $\hat{n}$  cannot be separated from 10–30-min rate modulation. The robust positive: fast cross-agent triggering,  $n_x \approx 0.07$  at 10–30 s lags, about 4× an agent-shift null ( $p = 2 \times 10^{-9}$ ). Humans and the nudger directly drive about 1.5% of events. **Round 1b** (corrected exogenous inputs, long days cut at operator-off gaps, day-edge-trimmed nulls): every period verdict is unchanged, fast cross-triggering survives the corrected null, and withdrawing all operator messages inside #51 (NE43) leaves  $\hat{n}$  and activity unchanged. No holdout run yet.

## PHYSICAL MODEL

Model 09 (Hawkes self-exciting point process). Events: agent messages; one realization per village day.  $\mu_B(t)$ : schedule baseline (per-day level  $\times$  30-min shape, plus a kickoff bump);  $g_{\text{exo}}$ : human and nudger input;  $\phi$ : triggering kernel;  $n$ : branching ratio, the mean number of events each event directly triggers (critical at  $n = 1$ ;  $\langle s \rangle$ : mean cascade size). Multivariate fit:  $\lambda_a$ , agent  $a$ 's intensity, with own and cross kernels;  $\bar{m}$ : active agents;  $w_m^x$ : cross weight at timescale  $\tau_m$ .

$$\begin{aligned}\lambda(t) &= \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j) \\ \phi(\tau) &= n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n) \\ \lambda_a(t) &= c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_x(t - t_j) \\ n_x &= (\bar{m} - 1) \sum_{\tau_m \leq 300\text{ s}} w_m^x\end{aligned}$$



**Observables.** (a) Branching ratio per goal period against swarm size, by coupling mode, with day-bootstrap 95% CIs: none is near critical, shared-objective periods sit no higher than free ones, and  $\hat{n}$  falls with size. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing: nearly every period sits above the diagonal, the one component that beats its strongest null.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A negative that matters a little: this swarm is nowhere near a runaway cascade, and neither more agents nor a shared goal pushes it there. Anyone worried about message storms gets mild reassurance, but activity is set mostly by the scheduler and roster, not by social contagion. The one robust number (about 0.07 extra messages from others within half a minute of each message) may just be agents waking when a message lands: a scaffold property. A naive constant-baseline Hawkes fit would have reported near-criticality. Practically: one pit-fall and one small coupling constant.

## RESULTS VS PREDICTIONS

| Prediction                                                                                        | Observed (round 1 → Verdict round 1b)                                                                     |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| P1 free weeks subcritical ( $\hat{n}_F < 0.5$ )                                                   | 0.53 → 0.52; $4/8 \geq 0.5$ ×                                                                             |
| P2 shared > free                                                                                  | 0.32 vs 0.53 → 0.29 vs 0.52 ( $p = \times 0.80$ )                                                         |
| P3 shared near critical ( $\geq 0.7$ )                                                            | 0.32 → 0.29 ×                                                                                             |
| P4 mode survives $N$ , $t = -0.19 \rightarrow -0.67$ ; $\log N t = \times$ hours, regime $-2.06$  |                                                                                                           |
| P5 #51 drifts toward $n = \rho = -0.64 \rightarrow -0.66$ (own × 1 split), $-0.70$ (shared units) |                                                                                                           |
| P6 fast $n_\times$ : shared > free                                                                | 0.075 vs 0.071 → 0.075 vs 0.073 ×                                                                         |
| Fast $n_\times$ vs agent-shift null                                                               | 0.070 vs 0.017 → 0.074 vs 0.011 ✓ after trimming ( $p = 2 \times 10^{-8}$ )                               |
| Native NE43: drive withdrawn                                                                      | $\hat{n}$ 0.44/0.35/0.47 (within 2 SD); ✓ talk rate $-11\%$                                               |
| Native #2 / NE14                                                                                  | $n_\times$ 0.16 vs null max 0.11; trim ✓/ $\sim$ moves $\Delta \leq 0.03$ , $\hat{n}_{\text{TALK}}$ rises |

## GOAL PERIODS IN DETAIL

G02 × 1b: failed : n TALK 0.72 → 0.75 on corrected inputs;...  
G03 ~ 1b: mixed : n TALK 0.66 → 0.66 on corrected inputs;...  
G04 × 1b: failed : n TALK 0.55 → 0.55 on corrected inputs;...  
G05 ✓ 1b: supported : n TALK 0.15 → 0.15 on corrected inputs;...  
G06 d 1b: descriptive : n TALK 0.65 → 0.32 on corrected inputs;...  
G07 ~ 1b: mixed : n TALK 0.47 → 0.47 on corrected inputs;...  
G08 × 1b: failed : n TALK 0.27 → 0.26 on corrected inputs;...  
G10 d 1b: descriptive : n TALK 0.22 → 0.22 on corrected inputs;...  
G11 ~ 1b: mixed : n TALK 0.61 → 0.60 on corrected inputs;...  
G12 d 1b: descriptive : n TALK 0.63 → 0.63 on corrected inputs;...  
+27 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 0 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

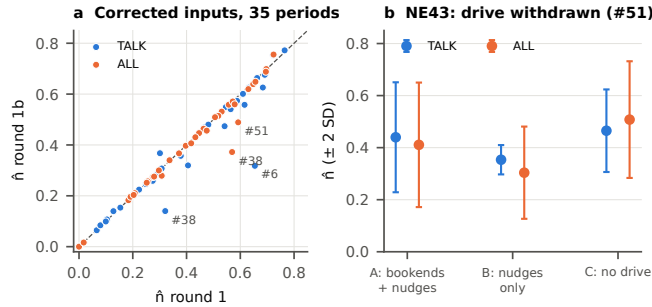
## CAVEATS

Pooled  $\hat{n}$  is still not identified against 10–30-min rate modulation. In regime III, `events_core` lacks computer-use turns, so ALL means something different. The M3 at-risk set ignores rooms. NE43's sides are confounded with the #focus room and side C has 5 days. Round-1b bootstraps use 50/25 replicates (round 1: 100/50). In #51,  $N$  is collinear with date and drive. H42 is fitting the read-out kernel this round does not model.

## ROUND 2

Leave the kernel to H42; next for H03: an autonomy fraction from the context ledger (H03-R1), NE10 (first nudges), and locking the holdout predictions (fast  $n_\times$  above the trimmed shift null;  $\hat{n} < 0.8$ ).

- **Essence**. How autonomous is the swarm: what fraction of its activity is self-generated rather than driven by the scaffold or humans?
- **H03-R1** Autonomy fraction: classify every turn's trigger (scheduled, external message, peer message, self-continuation) from the context ledger; autonomy = peer + self share, tracked across regimes.
- **H03-R2** The swarm dies without drive: during outages and nudger-off sessions, peer-driven activity decays with a measurable lifetime.
- **H03-R3** Criticality, if anywhere, lives in content cascades (H34), not in timing.



**Second figure.** (a) Per-period  $\hat{n}$  in round 1 vs round 1b: corrected exogenous drive and day splitting move almost nothing; #6 and #38 change because a long operator-off gap is now cut out of one day. (b) NE43 inside #51 at a fixed roster:  $\hat{n}$  does not fall when the daily bookends stop (B) or when the nudger stops (C).